Entropic Risk Measure

For a loss L and fixed $\gamma > 0$, define

$$\rho(L) = \frac{1}{\gamma} \ln \mathbb{E}[e^{\gamma L}].$$

We consider losses for which this expectation is finite.

1. Cash-additivity and monotonicity

For any constant $c \in \mathbb{R}$,

$$\rho(L + c) = \frac{1}{\gamma} \ln \left(e^{\gamma c} \mathbb{E}[e^{\gamma L}] \right) = c + \frac{1}{\gamma} \ln \mathbb{E}[e^{\gamma L}] = \boxed{\rho(L) + c.}$$

Thus adding a certain loss c increases the required capital by c. Equivalently, adding cash c reduces the loss to $L - c$ and gives $\rho(L - c) = \rho(L) - c$.

If $L_1 \leq L_2$ almost surely, then $e^{\gamma L_1} \leq e^{\gamma L_2}$ almost surely. Taking expectations and using that ln is increasing and $\gamma > 0$ gives

$$\boxed{\rho(L_1) \leq \rho(L_2).}$$

2. Convexity using Hölder's inequality

Let $0 < \theta < 1$, and set $p = 1/\theta$ and $q = 1/(1 - \theta)$, so $1/p + 1/q = 1$. Hölder's inequality gives

$$\begin{aligned} \mathbb{E}\left[e^{\gamma(\theta L_1 + (1-\theta)L_2)} \right] &= \mathbb{E}\left[(e^{\gamma L_1})^{\theta} (e^{\gamma L_2})^{1-\theta} \right] \\ &\leq \left(\mathbb{E}[e^{\gamma L_1}] \right)^{\theta} \left(\mathbb{E}[e^{\gamma L_2}] \right)^{1-\theta}. \end{aligned}$$

Taking logarithms and dividing by γ yields

$$\boxed{\rho(\theta L_1 + (1 - \theta)L_2) \leq \theta \rho(L_1) + (1 - \theta)\rho(L_2).}$$

The cases $\theta = 0$ and $\theta = 1$ are equalities. Together with Question 1, this proves that ρ is a convex risk measure.

3. Failure of coherence

Let $L \sim N(0, 1)$ and $\lambda > 0$. Completing the square gives

$$\begin{aligned} \mathbb{E}[e^{\gamma \lambda L}] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\gamma \lambda x - x^2/2} \, dx \\ &= e^{\gamma^2 \lambda^2 / 2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x - \gamma\lambda)^2/2} \, dx = e^{\gamma^2 \lambda^2 / 2}. \end{aligned}$$

Therefore

$$\boxed{\rho(\lambda L) = \frac{\gamma \lambda^2}{2},} \qquad \lambda \rho(L) = \frac{\gamma \lambda}{2}.$$

For example, $\rho(2L) = 2\gamma$ whereas $2\rho(L) = \gamma$. Positive homogeneity fails, so the entropic risk measure is **not coherent**.